

Yuting Zhou

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EDUCATION

Rutgers University , New Jersey, United States	Sep 2024 - Expected Jun 2026
• Major: M.S. Candidate in Computer Science	
Shanghai Institute of Technology , Shanghai, China	Sep 2020 - Jun 2024
• Major: Bachelor of Engineering in Software Engineering (GPA: 3.7, Top 10%)	

CERTIFICATES

- Alibaba Cloud Certified Associate - Big Data, Huawei Certified ICT Associate - Intelligent Computing

RESEARCH EXPERIENCE

Research Assistant <i>Hierarchical Text Classification, Data Engineering, Large Language Models</i>	Jun 2025 - Present
Rutgers Center for Advanced Infrastructure and Transportation (CAIT)	New Jersey, United States
• Architected scalable data engineering pipelines to merge and reconcile 100k+ multi-source railway incident and delay records, implementing automated deduplication, hierarchical label dictionary construction, and cross-dataset consistency validation.	
• Developed analytical and visualization tools to convert raw operational logs into structured daily, route-level, and location-based performance metrics, supporting delay pattern analysis and anomaly detection.	
• Designed and fine-tuned hierarchical multi-label classifiers based on pretrained transformer models (e.g., DeBERTa), modeling structured dependencies with supervised fine-tuning, encoder freezing strategies, and dependency-aware evaluation.	

INTERNSHIP EXPERIENCE

Front-end Engineer <i>JavaScript, Vue, React, WebSocket, MongoDB</i>	Jun 2023 - Aug 2023
Shanghai Tenchii Digital Tech Co., Ltd. (Tencent and Shanghai Metro Joint Venture)	Shanghai, China
• Optimized front-end performance of the Shanghai Transit App by refactoring component rendering and request handling logic, reducing page load latency by 30% and improving user retention.	
• Developed a real-time lottery system supporting high-concurrency traffic using WebSocket and MongoDB, ensuring low-latency updates and stable performance under peak load.	
• Designed and implemented 3 cross-team prototype interfaces in collaboration with product managers, translating Axure mockups into production-ready Vue components.	
Software Test Engineer <i>Pytest, JUnit, Selenium</i>	Jun 2022 - Sep 2022
Shanghai Spacecraft Electromechanical Equipment Co., Ltd.	Shanghai, China
• Built automated test environment across hardware configs with multi cases using Pytest/JUnit and CI pipeline.	
• Detected high-priority defects via Selenium UI automation, reducing production risk.	

PROJECT EXPERIENCE

Railway Incident Classification System <i>Hierarchical Multi-label Learning, DeBERTa-v3, Large-scale Log Mining</i>	Aug 2025 - Present
• Designed and trained a hierarchical multi-label classifier based on DeBERTa-v3 to model structured dependencies on real-world railway incident logs.	
• Built an end-to-end training and inference pipeline with supervised fine-tuning, warm-up scheduling, encoder freezing, and reproducible experiment management.	
• Engineered scalable data preprocessing workflows for multi-source log merging, deduplication, hierarchical dictionary construction, and label consolidation across 3-year datasets.	
• Evaluated structured predictions using hierarchical exact match, per-level F1, and dependency-aware consistency checks.	
AyeNote Cloud Notes Platform <i>Vue 3, Django, BeautifulSoup, Redis</i>	May 2022 - Apr 2023
github.com/Magicherry/AyeNote	
• Built an async crawler processing 5k+ pages/day at 99% uptime; boosted parsing accuracy from 56% to 78% using async I/O and SimHash.	
• Integrated LLM-based summarization to reduce per-note manual curation time by 60%, significantly improving content throughput and usability.	
• Implemented inverted indexing and keyword ranking to enable sub-second full-text search across 30k+ structured notes.	
• Developed real-time collaborative editing for 20+ concurrent users with <250ms latency using Django Channels and WebSocket.	
King of Bots <i>Vue 3, Spring Boot, MySQL, WebSocket, Redis</i>	May 2022 - Nov 2022
github.com/Magicherry/King-of-Bots	
• Developed a full-stack real-time battle platform supporting 50+ concurrent matches with <250ms end-to-end latency.	
• Architected distributed session and matchmaking management using Redis caching and pub/sub to ensure state consistency under high concurrency.	
• Optimized game state synchronization and WebSocket multiplexing, increasing throughput by 35% under load testing.	
• Designed and implemented 10+ secure REST APIs with JWT authentication and role-based access control (RBAC).	

TECH STACK & LANGUAGES

Programming	Python, C++, Java, Go
ML & NLP	PyTorch, Transformers, scikit-learn, TensorFlow, DeBERTa / BERT, LLM Fine-tuning, Hierarchical Classification
ML Systems	Mixed Precision Training, Gradient Accumulation, Multi-GPU / MPS Training, Experiment Tracking
Data & Analysis	NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering
Databases	SQLite, MySQL, Redis, MongoDB
Backend	Django, FastAPI, Spring Boot, RESTful APIs, Microservices
Frontend	HTML, CSS, JavaScript, Vue 3, React, Next.js
DevOps	Linux, Docker, Kubernetes, GitHub Actions
Languages	English, Mandarin Chinese (Bilingual)